

Student
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Total Points
98 / 100 pts

Question 1

150 / 50 pts

✓ - 0 pts correct

Question 2

248 / 50 pts

- 0 pts Correct

✓ - 2 pts PV = mZRT/M, so you should also include the molecular weight in the calculation. The remaining procedure is correct.

- 20 pts incomplete

- 25 pts late

No questions assigned to the following page.

$$\dot{Q}_{cv} = 0.4 T_b - 118.4$$

$$\dot{Q}_R = \epsilon \cdot \sigma \cdot A (T_b^4 - T_F^4)$$

$$= 0.4 \cdot (5.67 \times 10^{-8}) \cdot (0.02) (T_b^4 - 296^4)$$

$$\dot{Q}_T = \dot{Q}_{cv} + \dot{Q}_R = (0.4 T_b - 118.4) + (4.536 \times 10^{-10}) (T_b^4 - 296^4) = 1000$$

use excel to solve (goal seek) $T_b = 11.06 \text{ K} = 833^\circ \text{C}$

Case Study #1

Nitrogen Gas initially at 100 kPa + 300 K $\xrightarrow{\text{comp.}}$ 2000 kPa

$$PV^{1.25} = C$$

$$PV = RT$$

$$2000(0.045) = \left(\frac{8.314}{28.01} \right) T_2$$

$$a.) \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{100}{300} = \frac{2000}{T_2}$$

$$\frac{100}{300} = \frac{2000}{T_2}$$

$$T_2 = 6000 \text{ K} \quad T_2 = 303 \text{ K}$$

$$W = P dV$$

$$PV = RT$$

$$(100) V_1 = \left(\frac{8.314}{28.01} \right) (300)$$

$$P = \frac{86.5}{(0.89)^{1.25}}$$

$$(100)(0.89)^{1.25} = C$$

$$V_1 = 0.89$$

$$C = 86.5$$

$$P_1 V_1 = P_2 V_2$$

$$(100)(0.89) = (2000) V_2$$

$$V_2 = 0.0445$$

$$W = \int_{-89}^{0.0445} \frac{86.5}{V^{1.25}} dV = \boxed{-398.9 \text{ kJ/mg}}$$

No questions assigned to the following page.

$$P_R = \frac{1.01 \text{ bar}}{37.7} = .027$$

$$Z = .99$$

so yes, good assumption

$$\text{at } -30^\circ\text{C}, T_R = 1.83$$

$$\text{b.) } P_R = \frac{1}{33.9} = .029 \quad \left. \begin{array}{l} 100 \text{ kPa} = 1 \text{ bar} \\ \cancel{2000 \text{ kPa}} = \cancel{20 \text{ bar}} \end{array} \right\}$$

$$T_R = \frac{300}{126} = 2.38$$

$$Z = .98$$

$$P \sim (1.98)RT$$

$$2000 (0.045) = (1.98) \left(\frac{8.314}{28.01} \right) T_2$$

$$T_2 = 309.4 \text{ K}$$

$$\cancel{W = \int P dV}$$

$$W = \int_{.89}^{.045} (1.98) \frac{86.9}{V^{1.23}} dV$$

$$W = -390.92 \text{ kJ/kg}$$

No questions assigned to the following page.

c.) Using AI:

$$T_2 = 546 \text{ K} \quad W = 365.13 \text{ kJ/kg}$$

$$w/z, \quad T_2 = 546 \text{ K} \quad W = 365.2 \text{ kJ}$$

d.) The AI got the exact same answer for both assuming ideal gas and ~~assum~~ using z-values for real gas. I also obtained a smaller temperature.

Reflection: If ideal gas is assumed for ammonia plants when it is not the case, then the compressor might either compress too much or too little in the plant. This can decrease its effectiveness, and in the long run make it cost more to operate than it should.

Case Study 2:

$$a.) PV = nRT \quad (2500)(50) = n \left(\frac{8.314}{1000} \right) (25+273)$$

$$50.5 \times 2.02 = \boxed{56.96 \text{ g}} \quad n = 50.5 \text{ mol}$$

$$= \boxed{101.92 \text{ g}}$$

$$b.) PV = ZnRT \quad \frac{25}{13} = P_R = 1.92 \quad \frac{298}{33.2} = 8.97 \approx T_R$$

$$Z = 1.06$$

$$(2500)(50) = n(1.06)(8.314)(298) \quad n = 47.6 \text{ mol} \quad 47.6 \times 2.02 = \boxed{96.15 \text{ g}}$$

~~6x~~

$$(20000)(50) = n(8.314)(298)$$

$$n = 403.6 \text{ mol} \quad 403.6 \times 2.02 = 815.3 \text{ g}$$

$$815.3 - 101.92 = \boxed{713.4 \text{ g of additional H}_2}$$

$$\frac{735.8 - 713.4}{713.4} \times 100 = \boxed{3.14\%}$$

$$(20000)(50) = n(0.98)(8.314)(298)$$

$$n = 411.9 \text{ mol} \quad 411.9 \times 2.02 = 831.95 \text{ g}$$

$$831.95 - 96.15 = \boxed{735.8 \text{ g}}$$

percent error



No questions assigned to the following page.

after warming to 40°C (313K)

a.) $P(50) = (403.6)(8.314)(313)$
 $P = 23532.2 \text{ kPa} = \boxed{235.32 \text{ bar}}$

b.) $P(50) = (0.98)(403.6)(8.314)(313)$
 $P = 23061.5 \text{ kPa} = \boxed{230.62 \text{ bar}}$

c.) $(2000)(50) = (0.98) n (8.314)(313)$
 $n = 392.12 \text{ mol}$
 $392.12 \times 2.02 = \boxed{792.1\text{g}}$

$$\dot{V} = \frac{0.5 \text{ kg}}{1\text{s}} \times \frac{60\text{s}}{1\text{min}} \times \frac{2\text{min}}{1} = \boxed{60000\text{g}}$$

This amount of hydrogen release is dangerous because it is very flammable, so if it were to catch fire there

would be disastrous consequences.

According to chat GPT (who was given an exact screenshot of the questions)

mass (ideal): 105.8 kg

mass (real): 106.034 kg

mass (added): 711.356 kg

mass (add w/ Z): 668.4 kg $\rightarrow$ 6.35%

~~max mass receiver 711.4 kg~~

~~max mass (real) @ 668.411 $\leftarrow$ 6.35%~~

P(ideal): 210 bar

P(real): 226 bar

safe mass: 724.3 kg

My results differed slightly from the AI's from our slightly different calculated Z value from calculating P_r and T_R differently. The AI also used a slightly different variation in the equations used. ~~and~~ The Ideal gas assumption didn't really cause any extremely large error, however, when calculating the pressure after the temperature

was warmed to 40°C , the difference between the pressure found assuming ideal gas versus using a z -value for a real gas was around 5 bar, which can be significant in some scenarios. By adding in safety control features and ~~the~~ measurement indicators to prevent the tank from nearing its max capacity point. This case goes to show how important it is to use correct eqns and numbers from scenario to scenario. It's important to keep track of changes in the real world environment so your calculations don't end up incorrect or off. Just using hypothetical perfect ideal gas doesn't always cut it.

